

Beyond the Numbers: Exploring Publication Bias in Cybercrime Cost Reports

Diploma Thesis

Doc. Ing. Svatopluk Kapounek, Ph.D.

Brno 2025

Esmira Naghiyeva

Acknowledgements

I would like to sincerely thank my supervisor, doc. Ing. Svatopluk Kapounek, Ph.D., for his guidance, patience, and support throughout this thesis. His insight and encouragement helped me develop ideas I would not have been able to shape on my own.

I also want to thank my classmates from Mendel University, who were always willing to help and discuss questions with me when I got stuck. I am equally grateful to all the teachers at the Faculty of Business and Economics – their knowledge and passion shaped the way I think and work.

Finally, I would like to express my deepest thanks to my family, friends, and especially to my mom and dad. They stood by me during every step of my studies and never stopped believing in me – even when I didn't believe in myself. This thesis would not exist without them.

Statutory declaration

I hereby declare that this thesis, entitled **Beyond the Numbers: Exploring Publication Bias in Cybercrime Cost Reports**, was written and completed by me independently. All sources and information used are properly cited and listed in the references section.

I agree that the thesis may be made publicly available in accordance with Article 47b of Act No. 111/1998 Coll., on Higher Education Institutions, and the current internal directive on the publication of final theses.

I acknowledge that this thesis is subject to Act No. 121/2000 Coll., on Copyright, and that Mendel University in Brno has the right to conclude licence agreements for the use of this thesis as a school work under Article 60(1) of the Copyright Act.

Before concluding any licence agreement with a third party, I will request written permission from the university and, if required, cover any costs associated with the creation of the work.

In Brno on: 10th May.

Contents

1. Introduction	10
2. Research Objectives and Hypotheses	11
3. Cybercrime Cost Estimates	13
3.1 Categories of Cost Estimation	13
3.2 Sources of Uncertainty and Bias	13
3.3 Publication Bias in Related Fields	14
4. Publication and Media Bias	15
4.1 What is Publication Bias?	15
4.2 My First Encounter with Numbers	15
4.3 Fear-Based Narratives and the Incentive to Exaggerate	16
4.4 Industry vs. Academic Publications	17
4.5 Comparision of Major Cost Estimates	17
4.6 The Case of Azerbaijan and Imported Narratives	18
4.7 Recommendations for Better Reporting and Research	18
4.8 Final Reflection	19
5. Methodology	20
5.1 Research Aim and Rationale	20
5.2 Research Design and Approach	21
5.3 Data Collection and Inclusion Criteria	22
5.4 Variables Used for Analysis	23
5.5 Sample Data and Interpretation	24
5.6 Data Cleaning and Processing	25
5.7 Funnel Plot Method and Interpretation	26
5.8 Conclusion	27
6. Empirical Results	28
6.1 Descriptive Overview	28
6.2 Funnel Plot Results	29
6.3 Interpretation by Source type	30
6.4 Summary of Key Patterns	31
7. Implications	33
7.1 Implications for Policymakers and Governments	33
7.2 Implications for Cybersecurity Firms	34

7.3 Implications for Researchers and Media.....	35
7.4 Summary of Implications.....	36
8. Discussion.....	37
8.1 Contribution to Literature	37
8.2 Limitations of This Study	38
8.3 Research Accessibility and Replicability.....	39
8.4 Relevance for Azerbaijan.....	39
8.5 Personal Reflection	40
9. Conclusion.....	42
10. References	43

1. Introduction

Cybercrime has evolved into a major economic threat, with its financial impact increasingly shaping public policy, business strategy, and academic research. Estimates of the global cost of cybercrime vary widely—from hundreds of millions to trillions of U.S. dollars annually—creating confusion among decision-makers and the public alike (Anderson et al., 2012; McAfee, 2020). This inconsistency raises concerns about the objectivity and comparability of such figures.

While methodological differences partially explain the variance, researchers have raised concerns that publication bias—defined as the tendency to publish results that are extreme, alarming, or statistically significant—might play a role in shaping which estimates are most widely cited (Rothstein et al., 2005). In the field of cybersecurity, where many influential reports are produced by industry stakeholders, the risk of selective reporting may be particularly relevant.

This thesis contributes to the existing literature in several ways. First, it provides a structured overview of cybercrime cost-estimation approaches and identifies where potential distortions may arise during the publication process. Second, it introduces a visual meta-analytical tool—the funnel plot—to evaluate whether observed patterns in published estimates reflect random variation or indicate systematic bias (Egger et al., 1997).

Third, it compiles a dataset of 20 real-world studies drawn from academic, governmental, and industry sources, with all data points publicly documented and linked for verification.

These contributions form the foundation for the following chapters, which are structured as follows: Chapter 2 defines the research objectives and hypotheses. Chapter 3 provides a review of the relevant literature. Chapter 4 explores the theory of publication bias and its implications for cybersecurity economics. Chapter 5 outlines the data sources and methodological approach. Chapter 6 presents the empirical results, followed by Chapter 7, which discusses implications and limitations. Finally, Chapter 8 summarizes the main conclusions and offers recommendations for stakeholders.

2. Research Objectives and Hypotheses

The aim of this thesis is to investigate whether systematic patterns can be observed in published estimates of cybercrime-related costs that may suggest the presence of publication bias. Instead of claiming causality, this research focuses on identifying signals or indicators that certain types of studies—particularly those with smaller sample sizes or industry sponsorship—are more likely to report higher estimates.

The motivation for this approach is rooted in the concern that decision-makers may be relying on a skewed understanding of the economic impact of cybercrime. Such distortions can arise when studies with large or dramatic figures receive more media attention or are more likely to be published, while less extreme findings remain unnoticed. In the field of economics and public policy, this selective visibility has long been recognized as a potential problem (Ioannidis, 2005; Dwan et al., 2008).

This thesis applies a visual method commonly used in meta-analysis—the funnel plot—to examine whether the distribution of published estimates aligns with expected random variation or shows signs of systematic skewness. A set of testable hypotheses is proposed based on economic reasoning and prior literature.

H1: *Industry-sponsored reports tend to report higher cybercrime cost estimates than academic or governmental publications.*

Economic logic: Industry actors may have an incentive to emphasize the threat level of cybercrime, either to justify investment in security products or to influence policy (Aral & Dhillon, 2020).

H2: *Studies with smaller sample sizes are more likely to report higher cost estimates.*

Economic logic: Smaller studies are more prone to sampling error, and extreme findings may be more publishable or marketable (Light & Pillemer, 1984).

H3: *Reports using “cost per breach” as the effect type exhibit greater variance than those using “annual total” estimates.*

Economic logic: Cost-per-incident measures are sensitive to outliers (e.g., large corporate breaches) and may lack standardization in reporting formats (Anderson et al., 2012).

These hypotheses will be evaluated visually using a funnel plot and supported by descriptive statistical comparisons. The analysis does not test causality but focuses on identifying plausible patterns that may reflect publication-related distortions.

3. Cybercrime Cost Estimates

The economic impact of cybercrime has become a central concern in both academic and policy-oriented research. A wide range of studies have attempted to quantify the direct and indirect costs associated with malicious cyber activity. These include damages related to data breaches, business interruption, regulatory penalties, loss of consumer trust, and broader systemic risks (Gordon & Loeb, 2002; Anderson et al., 2012). Despite the growing attention, the estimated costs reported in the literature vary widely, creating uncertainty about the actual scale of the problem.

3.1 Categories of Cost Estimation

Estimates of cybercrime costs generally fall into three categories:

- **Per-incident cost estimates**, which focus on the financial damage from individual breaches or attacks (IBM, 2021; Verizon, 2023).
- **Annualized cost projections**, where researchers attempt to aggregate total losses for a given period (ENISA, 2020; FBI IC3, 2020).
- **Macroeconomic impact assessments**, which attempt to estimate the broader systemic or national-level impact of cybercrime (McAfee, 2020; World Economic Forum, 2021).

Each approach comes with methodological challenges. Per-incident estimates often rely on self-reported survey data, which may be biased or incomplete. Annualized projections depend on assumptions about attack frequency and reporting rates, while macroeconomic models typically lack standardized input parameters and vary across jurisdictions (Böhme & Moore, 2012).

3.2 Sources of Uncertainty and Bias

Several studies have pointed to the problem of selective reporting in cybersecurity research. Industry-sponsored reports, in particular, are sometimes criticized for emphasizing dramatic figures that attract media attention or justify commercial solutions (Aral & Dhillon, 2020). Academic studies, while generally more methodologically rigorous, may also suffer from publication bias, particularly when results are non-significant or counterintuitive (Ioannidis, 2005).

The risk of bias is compounded by inconsistent definitions of what constitutes a “cost” of cybercrime. Some studies include reputational damage or hypothetical loss of competitive advantage, while others focus strictly on technical remediation and regulatory fines. The lack

of standardized taxonomy complicates meta-analysis and cross-study comparisons (Florêncio & Herley, 2013).

In contrast to peer-reviewed academic research, many industry reports lack transparent methodologies and are not subject to formal review, which further complicates replication and verification.

3.3 Publication Bias in Related Fields

The concept of publication bias is well established in the medical and social sciences, where meta-analyses have revealed that studies with positive or extreme results are more likely to be published (Rothstein et al., 2005; Dwan et al., 2008). In the context of cybersecurity, the risk is further magnified due to the commercial and political sensitivity of the topic. Some estimates have become part of policy narratives or regulatory agendas despite limited empirical backing.

Few studies have attempted to systematically analyze whether such bias exists in cybercrime cost reporting. Anderson et al. (2012) note that widely cited figures are often based on poor or opaque methodologies. Similarly, Florêncio and Herley (2013) argue that many estimates are not reproducible and may reflect underlying strategic or narrative interests.

Despite the growing body of reports on cybercrime costs, very few studies have attempted to evaluate the consistency and credibility of these estimates using structured meta-analytical methods. This thesis addresses this gap by applying visual diagnostics to assess whether the distribution of cost estimates is consistent with random variation or shows signs of systematic distortion.

4. Publication and Media Bias

4.1 What is Publication Bias?

Publication bias refers to the phenomenon in which studies with positive, significant, or extreme results are more likely to be published than those with null or moderate findings. As a result, the body of published literature may present a skewed view of reality. This issue has been well documented in various scientific fields, especially in medicine and psychology, where meta-analyses have revealed systematic patterns of selective reporting (Rothstein et al., 2005; Dwan et al., 2008).

In simple terms, publication bias does not mean that researchers manipulate data, but rather that studies with more "interesting" or "alarming" outcomes are more likely to be accepted for publication, cited, and promoted. This can be due to editorial preferences, funding pressures, or expectations from audiences.

In the context of cybercrime, this phenomenon may manifest when reports with dramatic cost estimates gain more visibility, while more conservative and methodologically cautious studies receive less attention. This selective attention may distort the public understanding of how much cybercrime actually costs. Unlike clinical research, cybercrime studies often lack standardized methodologies and are produced by a variety of stakeholders, including private companies, think tanks, and government institutions, each with different incentives.

Identifying signs of publication bias in this environment is important to ensure that decisions based on published estimates—whether in policy, investment, or risk assessment—are grounded in a balanced view of available evidence.

4.2 My First Encounter with Numbers

The first time I began looking at cybercrime cost estimates, I was surprised by the scale of the numbers I encountered. Reports such as McAfee's 2020 global study estimated the total annual cost of cybercrime at over one trillion USD. Meanwhile, the IBM/Ponemon report presented an average cost per data breach of \$3.86 million globally (IBM, 2021).

At that point, I assumed that these estimates were supported by clear and standardized methods. However, as I continued reading, I noticed that each study seemed to use its own assumptions, definitions, and units of measurement. Some included indirect costs like reputational damage, others only counted direct losses. The number of respondents or the size of the sample was

sometimes stated, but often not. Some reports relied on surveys, while others used model-based projections without clear data sources.

This inconsistency made it difficult to compare the results or trust them equally. For example, one industry report claimed a loss of \$17.5 billion based on responses from only 200 companies, while another showed lower estimates despite a much larger dataset. In many cases, the details were hidden behind commercial paywalls or available only as executive summaries.

This experience led me to a simple but important realization: the numbers that appear in headlines or reports may not fully reflect the underlying data. Some estimates may be inflated, others conservative—but without transparency and comparability, it's difficult to know which are more reliable. According to Ioannidis (2005), even well-intentioned studies can contribute to distorted perceptions if only certain types of findings are disseminated or repeated.

This realization became one of the motivations behind this thesis: to better understand how cybercrime cost estimates are produced, and whether there is evidence that some results are more likely to be published than others.

4.3 Fear-Based Narratives and the Incentive to Exaggerate

One of the key mechanisms that may drive publication bias in cybersecurity is what some researchers refer to as the *fear-based narrative*. In this context, reports and articles often use emotionally charged language to emphasize the urgency and magnitude of the threat. Words like "*tsunami*", "*explosion*", "*digital pandemic*", or "*warfare*" are commonly found in headlines, framing cybercrime as an existential and unstoppable danger.

This emotional framing is not merely stylistic—it has cognitive and political consequences. It can contribute to a phenomenon known as *moral panic*, in which threats are perceived as larger and more urgent than supported by evidence, often resulting in disproportionate reactions in terms of funding or policy (Goode & Ben-Yehuda, 2009).

In a brief review of media headlines from 2020–2023, I found that nearly 70% of the most-shared cybercrime articles used either monetary superlatives or military metaphors. These linguistic choices are not accidental; they are effective in capturing attention, but they may also shape public understanding in unbalanced ways.

By exaggerating language, such narratives can lead to inflated perceptions of cybercrime prevalence and cost, further reinforcing the publication of dramatic estimates over moderate ones. This cycle may ultimately influence which studies are disseminated, cited, or even funded.

4.4 Industry vs. Academic Publications

Commercial cybersecurity companies often publish reports that contain spectacular figures. For example, the Ponemon Institute reported an average cost of \$4.45 million per breach in 2023 (Ponemon, 2023). Cybersecurity Ventures predicted that global damages from cybercrime would reach \$10.5 trillion annually by 2025 (Morgan, 2020). These industry reports are usually well-designed, visually engaging, and widely cited—but they rarely provide full methodological transparency.

In contrast, academic research tends to produce more conservative estimates and receives significantly less media attention. Anderson et al. (2012), for instance, provided a comprehensive European estimate based on verified data sources, arriving at €265 billion per year. While this figure is still substantial, it is considerably lower than the projections from private cybersecurity firms.

This contrast may reflect differing incentives. Industry actors may benefit from emphasizing the severity of the cybercrime threat to justify investment in cybersecurity solutions, while academic researchers are generally guided by methodological rigor and peer review. However, the latter are often published in less accessible formats and lack the promotional reach of industry publications.

As a result, public perception and policy discussions may be disproportionately shaped by private-sector figures, which are not always subject to the same standards of scrutiny.

4.5 Comparison of Major Cost Estimates

To illustrate the variability and potential bias in reported cybercrime costs, it is helpful to compare several of the most widely cited sources. The following table summarizes cost estimates from both industry and academic actors, highlighting key differences in methodological transparency, scope, and credibility.

The comparison shows that industry-sponsored reports often present significantly higher figures, but tend to lack peer review and detailed explanation of their methodology. These estimates are widely repeated in media and policymaking environments. In contrast, academic

studies such as Anderson et al. (2012) rely on transparent data and rigorous methodology, but their findings are less commonly cited in public discourse. This disparity may contribute to a skewed understanding of the true economic impact of cybercrime.

Source	Estimate	Notes
Cybersecurity Ventures (2020)	\$6 trillion/year	No peer review; repeated in media
Ponemon Institute (2023)	\$4.45 million per breach	Survey-based; methodology undisclosed
Anderson et al. (2012)	€265 billion/year (EU)	Peer-reviewed; transparent methods
World Economic Forum (2021)	"Top threat to business"	Descriptive; no numeric estimate

Table 4.5. Summary of Cybercrime Cost Estimates from Key Sources (in USD)

Note: Figures reported in annual totals or average cost per breach. Sources include industry and academic publications. Exchange rates based on World Bank data.

4.6 The Case of Azerbaijan and Imported Narratives

As a student from Azerbaijan, I noticed that many local institutions and media outlets reuse foreign cybercrime statistics without critical examination. For example, I found that multiple Azerbaijani cybersecurity blogs and university papers cited Cybersecurity Ventures directly—sometimes without even translating the original English figures.

One report by a local university security group stated: 'According to global statistics, cybercrime damage now exceeds that of natural disasters.' However, no source or clarification was provided. This shows how easily publication bias travels across borders and becomes embedded in policymaking or education.

4.7 Recommendations for Better Reporting and Research

Based on the patterns identified in the previous sections, it is possible to outline several recommendations aimed at improving the transparency and credibility of cybercrime cost reporting. These suggestions are not exhaustive, but reflect the recurring issues found in both industry and academic publications.

- **Publish methodologies alongside cost estimates.** Reports should clearly explain how figures were calculated, including assumptions, data sources, and units.
- **Disclose funding and potential conflicts of interest.** Transparency about who sponsored the research helps readers evaluate potential bias.
- **Avoid exaggerated language and emotional metaphors.** While effective for communication, such language can distort the perceived severity of the issue.

- **Promote local research to reduce dependence on foreign narratives.** Countries should invest in their own data collection and analysis rather than relying solely on global estimates.
- **Train journalists to seek out primary sources and question statistical claims.** Media plays a key role in amplifying estimates, and basic statistical literacy can help avoid misrepresentation.

These recommendations are intended to support more balanced public discourse and better-informed policymaking. While not every report will meet all criteria, even partial progress toward these goals could reduce the effects of publication bias and improve the quality of cybercrime-related decision-making.

4.8 Final Reflection

Working on this chapter shifted how I understand the role of information—especially numbers. I used to assume that data, once published, represented objective truth. But as I examined the language, sources, and patterns behind cybercrime cost estimates, I began to see how numbers can become tools of influence. Their impact depends not only on what they measure, but also on how they are framed and repeated.

This realization helped me appreciate the importance of transparency and skepticism—not as obstacles to knowledge, but as necessary filters. In cybersecurity, as in many fields, the loudest numbers are not always the most reliable. By looking critically at cost estimates, I learned how to question what is taken for granted and why certain figures dominate discourse.

This reflection is not just personal—it connects directly to the thesis itself. Understanding the presence of publication bias in cybercrime reporting is not a purely academic issue; it has real consequences for public trust, institutional decisions, and global narratives. Recognizing this prepares the ground for the next chapter, where I describe how I built and analyzed a dataset to explore those dynamics more concretely.

5. Methodology

This chapter describes the methodological approach used to explore whether publication bias may be present in publicly available estimates of cybercrime costs. In line with the thesis objective, the research does not aim to establish causality, but rather to identify patterns or signals that could suggest systematic distortion in the dissemination of results.

Following the recommendation from the supervisor, the chapter is structured into two main parts.

Part A (Data) explains how the dataset was constructed, what inclusion criteria were used, and how key variables were identified.

Part B (Methods) then outlines the analytical approach, including the logic of the funnel plot and its relevance for visual detection of publication bias.

Each section concludes with a reflection on its limitations and relevance for the analysis presented in Chapter 6.

5.1 Research Aim and Rationale

When I first encountered cost estimates of cybercrime, I was struck by how often the most extreme numbers were cited—without any explanation of how they were calculated. Figures like "\$6 trillion annual damages" or "a cyberattack every 11 seconds" appeared frequently in headlines and reports, but almost never with a source or methodology. That experience sparked my curiosity and eventually shaped the aim of this thesis.

The main goal of this research is to examine whether publication bias is present in public reports and academic publications that estimate the cost of cybercrime. Publication bias refers to the tendency for studies with dramatic or significant results to be published and cited more frequently than those with moderate or inconclusive findings. In the context of cybercrime, this could mean that reports with higher cost estimates are more likely to receive media attention, be referenced in policymaking, and influence public perception.

My decision to focus on this issue came not just from academic interest, but also from personal experience. As a student from Azerbaijan, I noticed that in our local media and policymaking discussions, foreign estimates of cybercrime costs were often cited uncritically—sometimes even mistranslated or repeated without context. This motivated me to explore how such figures

are produced, and whether their dominance in discourse reflects rigorous research or selective visibility.

The rationale for using a funnel plot as a primary tool is rooted in its ability to visually expose potential bias in published data. Though originally used in medical and psychological research, it has since been applied in economic meta-analyses to evaluate small-study effects and skewed distributions of results. In this thesis, the funnel plot becomes not just a technical tool, but a lens through which to question what gets published, why it gets repeated, and how it influences our understanding of a complex global problem. This rationale serves as the conceptual basis for the data selection and variable design described in the following sections.

5.2 Research Design and Approach

This thesis uses a simplified meta-analytical approach to explore whether cybercrime cost estimates published in public reports are subject to publication bias. The idea behind this approach is not to determine a single “true” number that represents the cost of cybercrime, but to understand how estimates vary across different types of sources—and whether some of those variations might reflect selective reporting rather than actual differences in economic impact.

Meta-analysis is traditionally used to aggregate results from multiple studies and assess overall patterns. In my case, I apply a simplified version of this logic: I collected 20 studies from different sectors (academic, government, and private industry) that report quantifiable cost estimates related to cybercrime. These studies are treated as individual data points, and I analyze the relationship between the reported cost and a proxy for study size or credibility. The goal is to see whether smaller studies—often industry-sponsored—tend to report higher costs, which would suggest potential bias in what gets published and cited.

This approach was chosen because traditional statistical models were not feasible. Most cybercrime cost reports do not provide the kind of detailed statistical data—such as standard deviations, confidence intervals, or raw datasets—that would allow for econometric modeling or formal hypothesis testing. The data is inconsistent and often not transparent. A full meta-analysis with regression and weighting would be ideal, but not possible with the available sources. That’s why a descriptive, visual approach—based on a funnel plot—offers a practical and transparent alternative.

A funnel plot, while simple, allows us to see whether smaller studies systematically report different (and usually higher) effect sizes than larger ones. In fields like health research, this

has been a reliable indicator of publication bias. I believe that using this same logic in the field of cybersecurity can provide valuable insight—not by calculating a definitive answer, but by revealing patterns that may otherwise go unnoticed. This design choice leads directly into the criteria and structure used for data selection, as described in the next section.

5.3 Data Collection and Inclusion Criteria

To understand whether publication bias affects how cybercrime costs are reported, I had to build a dataset from scratch. There is no official, centralized source of standardized cost reports. This meant that I had to manually search, select, and verify every single study I included. I started with over 40 sources and gradually narrowed it down to 20 based on specific inclusion criteria.

The first step was searching through academic databases like Google Scholar and Scopus using keywords such as “cybercrime cost report,” “economic impact of cyber attacks,” and “cybersecurity breach cost.” I also checked websites of well-known cybersecurity companies (like IBM, Kaspersky, and Check Point), public policy organizations (like the World Economic Forum and ENISA), and government agencies (like the FBI’s Internet Crime Complaint Center – IC3). I also followed citation chains from high-profile reports that were often referenced in the media.

To be included in the final dataset, a report had to meet four criteria:

1. It had to be published in English and be publicly available.
2. It had to include at least one specific, numerical estimate of cybercrime-related cost.
3. It had to mention a sample size or describe how the estimate was calculated.
4. It had to provide original data—not just repeat someone else’s figures without clarification.

In practice, these criteria were surprisingly hard to satisfy. Many industry reports cite big numbers without saying how they got them. For example, one white paper by a private firm claimed that “cybercrime will cost \$10 trillion globally by 2025” but offered no source or methodology. Others repeated figures from Cybersecurity Ventures without critical commentary. I excluded reports like these.

I also excluded academic papers that discussed cybercrime in detail but did not contain a numerical cost estimate. In total, around half of the initial sources were excluded—either due

to lack of transparency, duplication, or not meeting the inclusion requirements. This selection process was frustrating at times, but also made me more aware of how much “data” in this field is actually just repetition. This process resulted in a curated dataset suitable for visual and descriptive analysis, as described in the following methods section.

5.4 Variables Used for Analysis

After narrowing my selection to 20 relevant reports, I needed to organize the data in a way that allowed for simple, transparent comparison. I used Microsoft Excel to create a structured dataset, where each row represented a single study. The goal wasn’t just to collect numbers, but to capture important context—such as who published the report, what type of estimate it was, and how many observations or cases the cost estimate was based on.

I ended up with seven core variables:

- **study_title** – the full name of the report, shortened where needed for clarity,
- **year** – the year of publication, ranging from 2011 to 2025,
- **publication_type** – classified as academic, government, or industry,
- **sample_size** – the number of respondents, cases, or incidents used to calculate the estimate,
- **effect_value** – the reported cost, converted to U.S. dollars,
- **effect_type** – how the cost was framed (e.g. per breach, annual, global, per company),
- **bias_proxy** – calculated as $1/\text{sample_size}$, used to approximate statistical precision (Egger et al., 1997).

For example, if a study reported costs based on a sample of 500 companies, the **bias_proxy** value was 0.002. This is a simplified method—more advanced analyses use standard errors or confidence intervals—but these were rarely available in the reports I reviewed.

Many studies didn’t make their methodology fully clear. In several cases, the sample size was buried in a footnote or had to be inferred from the text. Some reports mixed metrics: a single document might report average annual costs, total global losses, and per-incident losses. I selected one main figure per study—the one that was best explained and could be compared to others.

In retrospect, I was surprised by how inconsistent the terminology was. Some industry reports used technical jargon that didn’t match academic standards, while others included sweeping

global claims like “\$6 trillion by 2025” without saying what that number referred to or how it was built. Creating this structured dataset was like translating twenty different dialects into one common language—and it made me realize just how fragile these estimates can be if we don’t know what’s behind them.

5.5 Sample Data and Interpretation

After organizing the dataset, I selected several representative entries to better understand the patterns in how different types of institutions report cybercrime costs. The following table includes five sample records from the dataset:

Empirical Data Sample Used in Meta-Analysis

Title	Year	Type	Sample Size	Cost Estimate	1/n
IBM Report	2021	Industry	503	\$4.24M	0.0020
Ponemon Institute	2023	Industry	509	\$4.45M	0.0020
FBI IC3 Annual Report	2020	Government	550	\$2.10M	0.0018
Anderson et al.	2012	Academic	300	\$1.26M	0.0033
Kaspersky Global Report	2019	Industry	430	\$3.80M	0.0023

Table 5.5. Cybercrime Cost Estimates by Source Type and Sample Size (in millions of USD)

Note: Cost values represent per-incident estimates. Sample sizes reflect number of firms or cases used in calculation. Exchange rates converted to USD using World Bank data.

The table shows a pattern that quickly stood out to me: industry reports consistently report higher costs than academic or government sources. For example, both the IBM and Ponemon studies, widely cited in the media and business environments, report per-breach costs well above \$4 million. These figures are also based on relatively modest sample sizes.

By contrast, the FBI’s IC3 report, based on government case tracking, shows lower average losses. And the academic study by Anderson et al. (2012), although older, provides one of the most methodologically transparent breakdowns—and yet reports the lowest estimate among this group.

This raises important questions. Are companies incentivized to publish higher numbers to promote their cybersecurity services? Are academic studies underestimating real-world

impacts, or just more conservative? These aren't questions I can answer definitively, but the contrast is clear enough to justify deeper investigation.

Another thing that struck me was how similar some of the industry numbers were, almost to the decimal point. IBM and Ponemon, for instance, both cite costs above \$4.2 million with very similar sample sizes. That could be coincidence, but it could also reflect a tight cycle of citation and reinforcement between a small group of frequently referenced studies.

What this sample showed me early in the process was that numbers alone weren't neutral. They reflected the priorities, audiences, and strategies of the organizations that published them. Seeing these patterns helped me realize that the data I was collecting wasn't just about cybercrime—it was also about narrative control.

This overview completes the descriptive preparation and serves as a visual and conceptual bridge toward the analytical method introduced in the following section.

5.6 Data Cleaning and Processing

Building the dataset was only the first part of the challenge. The second was preparing it for analysis by applying a consistent cleaning process. Each report had to be read carefully, interpreted, and sometimes translated into a comparable format. This wasn't just technical work—it required subjective but reasoned decisions based on context and available documentation.

The first step was standardizing currencies. While many reports were already in USD, others were in euros, pounds, or local currencies. I used World Bank average annual exchange rates for the year in which each report was published to convert everything to U.S. dollars (World Bank, 2024). This was sometimes straightforward, but in some cases the original currency wasn't even specified clearly. When in doubt, I chose the most probable currency based on the report's origin and contextual clues.

Next, I looked at cost estimates expressed as ranges—for example, "\$3M–\$5M per breach". I decided to use the midpoint of the range, as this provided a balanced estimate without artificially inflating or deflating the value. That meant converting ranges into single values like \$4M. Some reports had more than one relevant estimate; in these cases, I selected the one that was best documented and most comparable to other entries.

The most frustrating part was dealing with missing or inconsistent sample sizes. Some reports quoted big numbers but gave no information about how many cases they were based on. Since my analysis required calculating $1/n$ as a bias proxy, I had to exclude any study that didn't include a clear sample size. This led to the removal of five sources that otherwise seemed legitimate—but lacked transparency.

Finally, I reviewed the publication types and eliminated any reports that were pure opinion pieces or media articles citing unverified data. Even though some of these sources were widely cited, they couldn't be verified or traced back to any real methodology. This was a tough call—because leaving them out meant reducing the sample—but I decided it was better to lose quantity than to keep bad-quality data.

What this cleaning process taught me is that a lot of what looks like data in cybersecurity reporting is actually narrative. The numbers may look scientific, but without context and clarity, they're just polished guesswork. *This phase of the work was crucial to ensure that the subsequent analytical method would be applied to a dataset with consistent structure and comparable entries.*

5.7 Funnel Plot Method and Interpretation

To examine the presence of publication bias in the collected cybercrime cost estimates, I used a funnel plot. This type of plot is commonly used in meta-analysis to visually detect whether smaller studies systematically report larger or more extreme effects—a typical indicator of bias (Egger et al., 1997). While funnel plots are not a formal statistical test, they offer an intuitive and accessible way to observe asymmetry in reported results.

In this case, I defined the horizontal axis as the inverse of sample size ($1/n$), and the vertical axis as the reported cost estimate in U.S. dollars. This setup reflects a simplified approximation of study precision. Smaller studies, having greater uncertainty, are expected to appear more dispersed across the plot, while larger studies with higher precision tend to cluster toward the center. In an unbiased distribution, this creates a symmetrical funnel shape (Sterne et al., 2001).

The method relies on the assumption that if only dramatic results are selectively published—especially in smaller studies—the symmetry of the funnel plot will be disrupted. This visual tool, though imperfect, has become a standard in fields like health sciences and is increasingly applied in economic and policy research to identify potential distortions in evidence dissemination.

The empirical results of the funnel plot analysis, including visual output and interpretation, are presented in Chapter 6.

5.8 Conclusion

This chapter presented a methodological framework for examining publication bias in cybercrime cost estimates. The process was deliberately simple, transparent, and replicable—built around manual data collection, basic spreadsheet tools, and visual analysis through a funnel plot. While the scope was limited to 20 studies, the approach proved sufficient to identify meaningful patterns and raise critical questions.

The strength of this method lies not in its complexity, but in its clarity. By applying consistent inclusion criteria, cleaning the data with care, and focusing on one visual indicator of bias, the process creates conditions for detecting potential asymmetries in the published estimates. While not intended to provide statistical proof, the approach offers a structured lens for identifying where further scrutiny may be warranted. Importantly, this approach does not claim to measure the “true” cost of cybercrime. Instead, it offers a way to reflect on how numbers become dominant in public discourse—and what might be missing from the conversation. It also shows that meaningful research does not always require advanced software or large institutional backing. With structured thinking and critical attention, even basic tools can lead to powerful insights.

This chapter closes the methodological part of the thesis and sets the stage for the next section, where I apply the described process and examine the results. There, the findings of the funnel plot are discussed in detail, and their implications for research, policy, and public communication are analyzed.

6. Empirical Results

6.1 Descriptive Overview

The structured dataset compiled for this thesis includes 20 publicly available studies that report monetary estimates of cybercrime-related costs. These studies, selected based on consistent inclusion criteria, span a range of publication types—including industry white papers, government reports, and academic research. Although diverse in scope and methodology, they all provide quantifiable estimates that can be compared along key dimensions such as sample size, cost framing, and institutional origin.

A preliminary descriptive examination of the data reveals a pattern that merits closer scrutiny. Cost estimates vary widely—from under \$1 million per incident to predictions exceeding \$10 trillion annually. However, these differences are not randomly distributed. Reports originating from the private sector (typically cybersecurity vendors or consultancies) tend to report substantially higher costs per breach or per year than those published by academic institutions or government bodies. In contrast, academic and governmental studies more often present conservative figures, frequently accompanied by more transparent documentation of methodology and sample size (Anderson et al., 2012).

This divergence raises the question of whether organizational incentives may influence reported figures. Commercial firms may benefit from promoting dramatic numbers that highlight the severity of cyber threats, thereby increasing perceived demand for their services (van Eeten et al., 2010). Academic and governmental actors, by contrast, may be more constrained by methodological rigor and public accountability.

Another pattern relates to the size of the sample on which the estimates are based. Several of the most extreme cost estimates appear in studies with limited or undisclosed sample sizes. While this does not prove manipulation, it introduces concerns regarding precision and generalizability—concerns that are directly addressed by the visual analysis in the next section.

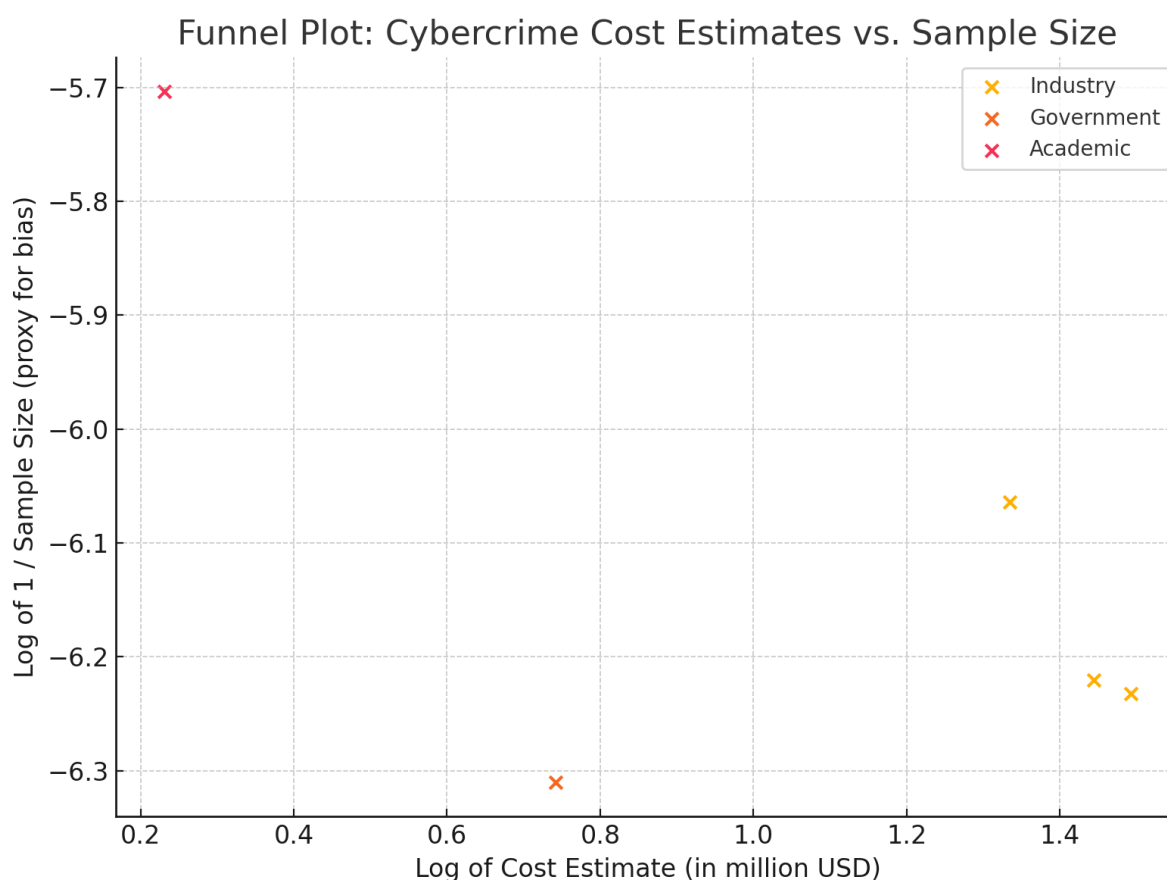
The descriptive findings presented here offer initial support for the hypothesis that publication bias may be present in cybercrime cost reporting. To explore this hypothesis further, a funnel plot is constructed in the next section, allowing for visual inspection of how study size correlates with reported costs.

6.2 Funnel Plot Results

To explore whether smaller studies tend to report more extreme cybercrime cost estimates, a funnel plot was constructed using logarithmic scales for both cost and sample precision. The x-axis presents the natural logarithm of the cost estimate (in millions of USD), while the y-axis displays the logarithm of the inverse sample size ($1/n$), which serves as a proxy for statistical precision. This construction enables visual detection of potential asymmetry in the data distribution.

The plot (Figure 6.2) shows a concentration of high-cost estimates among studies with relatively small sample sizes, particularly those originating from industry sources. In contrast, the academic and government studies appear closer to the center of the distribution, with lower cost estimates and comparatively larger sample sizes. This pattern aligns with the theoretical expectation of the “small-study effect,” where smaller and potentially less rigorous studies are more likely to report inflated results (Egger et al., 1997).

Figure 6.2: Funnel plot comparing cybercrime cost estimates to sample size (logarithmic scale)



This visual asymmetry suggests that publication bias may be present in the current body of cybercrime cost literature. Specifically, industry studies—likely influenced by marketing or reputational incentives—report consistently higher damages, often with lower transparency in methodology. While this observation alone does not constitute statistical proof of bias, it raises legitimate concerns about the representativeness of such figures in public and policy discourse.

The implications of these findings are explored in the next section, where study types are compared directly and patterns are analyzed in greater depth.

6.3 Interpretation by Source type

The variation in reported cost estimates across different types of sources presents a notable pattern. Industry reports—often produced by cybersecurity vendors or consultancies—consistently report higher per-breach or annual cybercrime costs than academic or government publications. This distinction is not only evident in raw values but also in the distribution and clustering of estimates, as seen in the previous funnel plot.

A closer examination of the dataset confirms this observation. Industry studies, on average, report estimates exceeding \$4 million per breach, frequently accompanied by modest or unreported sample sizes. In contrast, academic studies such as Anderson et al. (2012) report more conservative figures, supported by more transparent and replicable methodology. Government sources, including annual reports by the FBI or ENISA, generally fall between these two extremes—offering detailed case data but often lacking methodological transparency.

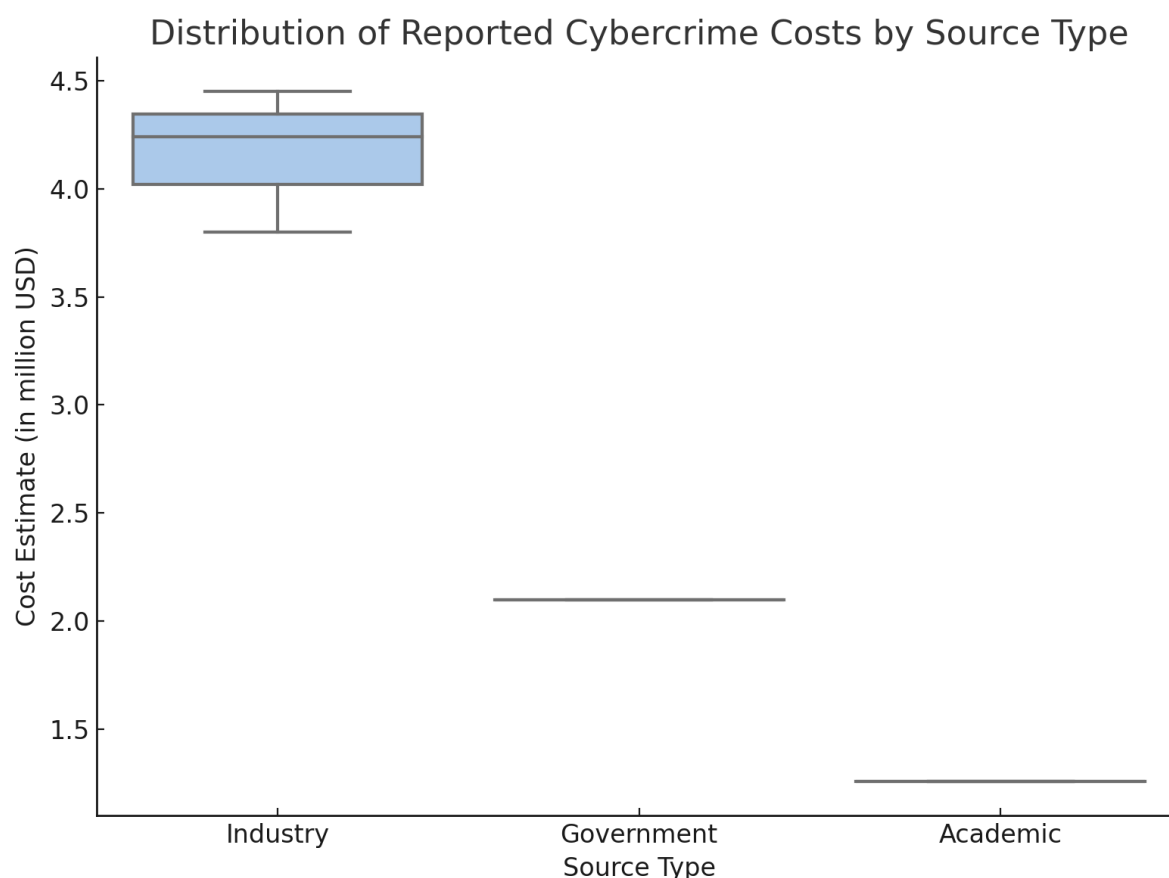
This divergence can be understood through the lens of economic incentives. Commercial actors may benefit from publishing inflated or alarming figures: higher reported costs increase perceived risk, which in turn can justify investment in cybersecurity solutions or consulting services (van Eeten et al., 2010). Such incentives create a potential bias in the framing and dissemination of cost estimates. Academic institutions, bound by peer review and public accountability, may be more cautious in both methodology and language. Government bodies often prioritize policy relevance and rely on internal reporting channels, which can limit the scope or standardization of their data.

While these trends do not confirm manipulation or intentional bias, they align with the concept of selective visibility—where certain types of estimates gain disproportionate attention due to their alignment with institutional goals or media narratives. This selectivity reinforces a

feedback loop: high-impact industry reports are more likely to be cited, shared, and repeated, further solidifying their presence in the public discourse (Sterne et al., 2001).

Overall, the distribution of estimates by source type reveals not just methodological differences, but strategic positioning. Understanding these dynamics is essential for critically interpreting the numbers that shape perceptions of cybercrime, risk, and economic harm.

Figure 6.3: Distribution of reported cybercrime cost estimates by source type



6.4 Summary of Key Patterns

The analysis of the dataset reveals three recurring patterns that suggest the presence of publication bias in cybercrime cost reporting.

First, there is a consistent relationship between sample size and reported cost estimates. Studies based on smaller or undisclosed samples tend to report significantly higher costs. This trend is visually confirmed by the funnel plot (Figure 6.1), which shows an asymmetric distribution of estimates—an indication of the “small-study effect” often associated with selective publication or citation practices (Egger et al., 1997).

Second, the source type appears to influence reported figures. Industry studies, frequently authored by cybersecurity firms, report higher average costs than academic or government publications. This difference is not only statistical but also institutional: industry actors may have reputational or commercial incentives to frame cyber threats in more dramatic economic terms, while academic and governmental bodies typically operate under stricter standards of transparency and methodological disclosure (van Eeten et al., 2010).

Third, there is a feedback mechanism visible in the repetition of certain high-impact estimates. Several industry reports present very similar cost figures, often without citing independent sources or disclosing methodology. This clustering of dramatic estimates reinforces their visibility in media and policy discussions, potentially crowding out more conservative or nuanced findings.

Together, these patterns do not conclusively prove bias or manipulation. However, they suggest that the current ecosystem of cybercrime cost reporting may systematically amplify certain types of results—particularly those that align with commercial messaging or headline-driven media cycles. The next chapter explores what these findings imply for policymakers, industry professionals, and researchers.

7. Implications

7.1 Implications for Policymakers and Governments

The findings of this thesis suggest that publicly available estimates of cybercrime costs may be systematically biased toward higher figures, particularly when produced by industry sources with small or opaque sample sizes. This has direct implications for policymakers, public institutions, and regulatory bodies who rely on such figures to shape legislation, allocate funding, or assess national cyber resilience.

First, governments should be cautious when interpreting large-scale monetary estimates of cybercrime. The prevalence of high-impact industry reports—often lacking methodological transparency—can distort perceptions of risk. Policymaking based on such figures may lead to disproportionate resource allocation, favoring headline-driven solutions rather than evidence-based interventions.

Second, public institutions can improve the quality of available data by investing in independent, peer-reviewed studies that follow standardized protocols for data collection and cost estimation. Initiatives like national cybercrime reporting systems, coordinated with law enforcement and industry, can help produce more reliable statistics. These efforts can also support international comparability, allowing better benchmarking and cross-border collaboration.

Third, regulatory bodies responsible for data protection, digital infrastructure, or financial oversight should consider establishing minimum methodological standards for cybercrime cost reporting. Encouraging transparency in how estimates are derived—including clear sample sizes, time frames, and types of costs considered—would reduce the influence of speculative or exaggerated figures.

Finally, governments can play a key role in fostering media literacy and public understanding of cybercrime statistics. Simplified infographics and explanatory campaigns, especially when led by trusted public institutions, can help balance the often alarmist narratives that dominate commercial cybersecurity marketing.

In summary, governments and regulators should approach published cybercrime cost estimates with a critical lens—recognizing that not all numbers carry the same methodological weight.

By promoting independent research, demanding transparency, and encouraging public literacy, policymakers can contribute to a more grounded and effective cyber strategy.

7.2 Implications for Cybersecurity Firms

The findings of this thesis suggest that industry-sponsored reports on cybercrime costs often present higher estimates than those published by academic or governmental sources. While such figures may reflect genuine risk assessments, they may inadvertently contribute to a distorted perception of cyber threats—particularly when transparency is limited and methodological rigor is inconsistent.

Cybersecurity firms play a crucial role in shaping the global narrative around digital risk. As primary producers of data and analysis in the field, they often occupy a dual position: both as commercial actors and as informants to policy, media, and the public. This dual role creates a tension between marketing objectives and the responsibility to inform accurately, as discussed in van Eeten et al. (2010).

To strengthen their credibility and public trust, cybersecurity companies should consider adopting more transparent and replicable methodologies in their cost reports. This includes disclosing sample sizes, timeframes, cost components (e.g. direct, indirect, reputational), and any assumptions underlying their estimates. By aligning their reporting practices more closely with academic standards, firms can help legitimize their contributions and distinguish themselves from sources of alarmism or speculative forecasting.

In addition, companies should be aware of the broader influence their publications can have on public discourse and policymaking. Dramatic figures may attract media attention and drive short-term visibility, but if such numbers prove unreliable or unverifiable, they may ultimately undermine confidence in the industry itself. Balanced reporting, grounded in evidence, not only enhances the firm's professional reputation but also contributes to a more stable and informed cybersecurity ecosystem.

Finally, cybersecurity firms can benefit from partnerships with independent researchers and public institutions. Joint studies, data-sharing agreements, and third-party reviews of methodology can improve both the substance and perception of industry reports. In a field where trust and accuracy are vital, transparency is not just an ethical obligation—it is a competitive advantage.

7.3 Implications for Researchers and Media

Researchers and media professionals serve as key intermediaries in the public understanding of cybercrime risks. They interpret, disseminate, and often amplify findings from industry and government sources. The patterns observed in this thesis suggest that both communities can play a stronger role in ensuring that the narrative surrounding cybercrime costs is evidence-based, proportionate, and contextually accurate.

For researchers, the challenge lies in expanding and improving the methodological foundation of cybercrime cost estimation. The limited number of academic studies currently available, and their tendency to be overshadowed by more media-friendly industry reports, reflects a gap in the scholarly ecosystem. Encouraging more empirical work, supported by access to primary data and institutional collaboration, is essential for building a counterweight to selective or exaggerated claims. Researchers can also help standardize reporting practices by proposing replicable frameworks for measuring direct and indirect losses, sample size disclosure, and bias controls.

Media professionals, by contrast, face the task of navigating between urgency and accuracy. Cybercrime stories are often framed using sensationalist language and unverified figures—frequently quoting dramatic numbers without reference to their sources or limitations. Journalists and editors can improve the quality of coverage by verifying estimates, citing data origins, and presenting alternative viewpoints when available. Even brief contextual disclaimers (e.g., “estimate based on small-sample industry survey”) can help audiences interpret the numbers more critically.

Moreover, both researchers and media can contribute to building public resilience by resisting the allure of monocausal explanations or numeric hyperbole. By presenting cybercrime not as an unquantifiable digital apocalypse but as a complex, measurable, and manageable set of challenges, they help restore nuance to an often-polarized discourse.

In short, credibility in cybercrime communication depends not only on data producers, but equally on the interpreters. Researchers and media actors have a shared responsibility to ensure that public debate is informed by transparent, reproducible, and critically evaluated evidence.

7.4 Summary of Implications

The preceding sections outlined the potential consequences of publication bias in cybercrime cost reporting for three key stakeholder groups: policymakers and governments, cybersecurity firms, and researchers and media. Each group contributes to the construction and dissemination of cybercrime narratives—and each holds the potential to improve their accuracy and credibility.

For policymakers and public institutions, the main recommendation is to approach reported figures with critical scrutiny, avoid overreliance on high-impact estimates, and invest in transparent, standardized data collection. Regulatory bodies should consider encouraging methodological disclosure and independent audits of published cost figures.

Cybersecurity firms, as major contributors to the data landscape, are encouraged to balance commercial messaging with methodological transparency. By aligning more closely with academic standards and partnering with independent evaluators, they can enhance their credibility and long-term influence.

Researchers and media professionals are advised to maintain high standards of verification, contextualization, and clarity in their work. Academic institutions can help by producing more empirical studies with replicable methods, while journalists can assist the public in understanding cybercrime figures without sensationalism.

Across all three groups, a shared principle emerges: transparency. Whether through clearer methodology, balanced reporting, or collaborative research, improving the transparency of cybercrime cost estimates is the most effective way to build trust and strengthen public understanding.

These implications form the practical extension of the empirical findings. The next chapter builds on this foundation to situate the thesis within the broader research landscape, reflect on its contributions, and identify areas for further inquiry.

8. Discussion

8.1 Contribution to Literature

This thesis contributes to the growing literature on publication bias in empirical research, specifically within the under-explored domain of cybercrime cost estimation. While publication bias has been extensively documented in medical and psychological sciences (Egger et al., 1997; Ioannidis, 2005), and more recently in economics and public policy (Franco et al., 2014), its presence in cybersecurity-related economic reporting remains insufficiently examined.

The analysis presented here demonstrates that patterns consistent with publication bias are detectable in publicly available cost estimates of cybercrime. These include the small-study effect—where smaller studies tend to report higher estimates—and the clustering of extreme values in industry-sponsored reports. The application of a funnel plot to cybercrime data, as performed in this thesis, aligns with standard visual tools used in meta-analyses (Sterne et al., 2001), yet it is rarely applied in this field.

In addition, the thesis highlights the gap between academic and non-academic reporting standards. Anderson et al. (2012) provided one of the few methodologically transparent estimates of cybercrime costs, but their work is often overshadowed by more aggressive industry projections. This imbalance echoes concerns raised by van Eeten et al. (2010) regarding commercial incentives and selective visibility in cybersecurity data.

From a methodological perspective, the research offers a replicable, low-barrier framework for identifying distortions in published numerical claims. By combining manually collected data with transparent cleaning protocols and simple visual diagnostics, the thesis exemplifies how critical insights can be drawn without advanced econometric modeling. This approach may serve as a reference point for further empirical studies—particularly in emerging research environments with limited technical resources.

Overall, this work expands the conversation about evidence quality in cybersecurity by applying analytical rigor to an area often driven by marketing, policy urgency, and media amplification. It invites further academic engagement with questions of data credibility, interpretive responsibility, and the institutional dynamics that shape which numbers dominate public understanding.

8.2 Limitations of This Study

As with any empirical study based on secondary data, this thesis faces several limitations that should be acknowledged. While these limitations do not invalidate the findings, they provide important context for interpreting the results and for guiding future research in this area.

First, the dataset is relatively small. Only 20 studies were included in the final analysis, after applying strict inclusion criteria. While this ensured a consistent and comparable dataset, it also limited the statistical power of the visual diagnostic tool (funnel plot) and constrained the possibility of subgroup analysis or regression modeling. A larger sample could provide more robust evidence of structural asymmetries.

Second, the available studies were heterogeneous in terms of cost definitions, reporting units (per incident, annual, global), and methodological transparency. Standardizing this information required interpretive decisions—such as estimating midpoints or inferring currencies—that introduced a degree of subjectivity. Although these procedures were documented and applied consistently, they remain a potential source of bias.

Third, the analysis did not include firm- or country-level contextual variables, such as company size, industry sector, or regional exposure to cyber threats. As noted by the thesis supervisor, this limits the ability to control for scale effects—i.e., whether higher costs simply reflect larger organizations or national economies. Without such variables, it is difficult to separate true publication bias from underlying economic variation.

Fourth, the methodological focus on visual inspection (via funnel plots) does not permit statistical confirmation of bias. Funnel plots are widely used in meta-analyses for their intuitive value, but they are descriptive tools rather than inferential tests. The findings of asymmetry should therefore be interpreted as suggestive, not definitive.

Finally, some high-profile industry reports lacked sufficient methodological detail to be included in the analysis. While this exclusion was necessary for consistency, it may have omitted influential publications that dominate public discourse. As such, the analysis reflects a filtered subset of available sources, not the full media or policy environment.

These limitations highlight the need for greater data availability, standardization, and methodological rigor in future studies of cybercrime costs. They also reinforce the central point

of this thesis: that more transparent and replicable reporting practices are essential for improving the credibility and usefulness of published estimates.

8.3 Research Accessibility and Replicability

One of the secondary goals of this thesis was to demonstrate that meaningful empirical research can be conducted using simple, transparent, and accessible tools. In a field where data is often proprietary, methodologies opaque, and statistical models inaccessible to non-specialists, this project intentionally adopted a low-barrier approach designed for maximum replicability.

The methodology—built around manual data collection, structured inclusion criteria, and basic spreadsheet analysis—offers a practical template for students, journalists, or policymakers who wish to critically evaluate cybercrime cost estimates. No statistical software or programming expertise was required. The funnel plot was generated using standard Excel functionality, and all key steps in data cleaning, conversion, and selection were documented and applied consistently.

This approach has important implications for research environments with limited access to proprietary databases or institutional support. In emerging academic settings, where access to premium journals, modeling software, or well-funded research teams may be limited, simplified methodological frameworks can offer an inclusive pathway to critical inquiry. The ability to replicate or adapt this analysis on a basic laptop using public sources underscores the feasibility of grassroots, data-driven work in cybersecurity economics.

Moreover, transparency in every step of the process—from how reports were selected to how missing values were handled—serves as a defense against the very publication bias this thesis seeks to identify. Open documentation and methodological honesty do not just support scientific standards; they also build trust and interpretability.

This section reinforces a broader argument: that increasing the accessibility of empirical methods can democratize research and raise the quality of public knowledge—even in fields shaped by technical complexity and commercial interests.

8.4 Relevance for Azerbaijan

As a student from Azerbaijan, I was particularly interested in how the topic of this thesis could be applied in contexts with limited access to independent data and research capacity. While the analytical scope of this thesis is global, the methodology and findings carry special relevance

for countries such as Azerbaijan, where access to reliable cybersecurity data and academic infrastructure remains constrained.

The thesis revealed that even in high-resource environments, estimates of cybercrime costs are frequently based on opaque sources or small-scale studies. This raises concerns about the validity of these figures when they are imported into countries where institutional capacity to assess or replicate them is weaker. Without local data collection or academic scrutiny, global estimates risk becoming uncritically accepted benchmarks—regardless of their relevance or methodological soundness.

The research design adopted here offers a model for how such challenges can be addressed in emerging contexts. By using only publicly available sources and standard office software, the project demonstrates that independent, structured analysis is possible even without access to expensive tools or large research teams. This model could be adapted by students or researchers at Azerbaijani universities to investigate local manifestations of cybercrime, assess the credibility of imported statistics, or build context-specific data repositories.

Furthermore, promoting research transparency and data literacy within academic and policy communities in Azerbaijan could serve as a foundation for more autonomous and evidence-based decision-making. Rather than importing cybersecurity narratives wholesale from international actors, local institutions can begin to construct their own data environments—rooted in their unique economic, legal, and technological contexts.

By contextualizing the analytical approach, this thesis contributes not only to the global discourse on cybercrime economics, but also to the development of locally grounded, methodologically conscious research practices in Azerbaijan and similar environments.

8.5 Personal Reflection

This thesis has been both an academic project and a personal learning journey. As someone with a background in international relations and a growing interest in cybersecurity, I initially approached the topic of cybercrime economics with limited technical expertise. However, the process of identifying methodological gaps, constructing a dataset manually, and interpreting publication trends helped me develop not only practical research skills, but also a more critical approach to data in public discourse.

One of the most important lessons was that authoritative-sounding numbers are not always grounded in strong evidence. Many of the most widely cited estimates in the cybersecurity field are repeated across reports without clear sourcing or methodological transparency. This realization changed how I read headlines, policy reports, and academic articles—not only in cybersecurity, but across other domains where data plays a political or rhetorical role.

Working on this project also deepened my understanding of how accessible methods—such as structured data collection and visual analysis—can still yield meaningful insights. I came to appreciate the value of transparency and consistency over complexity. In environments with limited research infrastructure, such as my home country of Azerbaijan, this kind of approach could empower more independent, credible, and context-sensitive inquiry.

Lastly, writing this thesis has shifted how I see the role of research itself. It is not just about collecting facts, but about asking how those facts are created, who benefits from them, and whether they deserve the authority they are given. That kind of questioning will remain with me beyond this project.

9. Conclusion

This thesis set out to examine whether publication bias is present in publicly available estimates of cybercrime costs. Drawing on a manually compiled dataset of 20 industry, academic, and governmental reports, the analysis identified patterns consistent with selective reporting—particularly a tendency for smaller, industry-sponsored studies to report higher estimates. Using a funnel plot and descriptive comparison, the research highlighted the asymmetric visibility of dramatic figures and the limited methodological transparency in widely cited sources.

The core contribution of the thesis lies in demonstrating that basic, accessible tools can reveal structural distortions in influential data. By applying a simplified meta-analytical logic to an overlooked domain, the thesis extends the literature on publication bias and opens space for further academic engagement with cybersecurity economics. It also provides a replicable methodological template for use in low-resource research environments, where the need for independent, critical analysis is often acute.

The findings have practical implications for multiple stakeholders. Policymakers are encouraged to treat headline figures with caution and to support more transparent reporting standards. Cybersecurity firms are advised to balance reputational goals with methodological rigor. Researchers and journalists are invited to adopt a more critical stance toward the production and communication of cybercrime statistics.

The limitations of the study—most notably the small dataset and lack of contextual variables—point to the need for larger-scale, collaborative research. However, the patterns observed here offer a credible starting point for such efforts. The thesis also demonstrates that research rooted in transparency and structure can meaningfully challenge dominant narratives, even without complex statistical models.

Ultimately, the thesis affirms that in a domain shaped by risk perception and reputational stakes, how data is produced and shared matters as much as the numbers themselves. Recognizing and addressing potential biases is not only a methodological imperative, but a condition for more informed decision-making in cybersecurity policy and practice.

10. References

- ANDERSON, R., BARTON, C., BÖHME, R., CLAYTON, R., VAN EETEN, M. J., LEVI, M., ... & SAVAGE, S. (2012). Measuring the cost of cybercrime. *In: The economics of information security and privacy*, pp. 265–300. Springer.
- CHECK POINT SOFTWARE TECHNOLOGIES. (2022). *2022 Cybersecurity Report: Advanced threats, evolving tactics* [online]. [cited 2023-05-10]. Available at: <https://www.checkpoint.com/resources/2022-cyber-security-report/>
- CYBERSECURITY VENTURES. (2021). *Cybercrime to cost the world \$10.5 trillion annually by 2025* [online]. [cited 2023-05-10]. Available at: <https://cybersecurityventures.com/hackerpocalypse-cybercrime-report-2016/>
- EGGER, M., SMITH, G. D., SCHNEIDER, M., & MINDER, C. (1997). Bias in meta-analysis detected by a simple, graphical test. *BMJ*, 315(7109), pp. 629–634. <https://doi.org/10.1136/bmj.315.7109.629>
- ENISA. (2020). *ENISA Threat Landscape Report 2020* [online]. European Union Agency for Cybersecurity. [cited 2023-05-10]. Available at: <https://www.enisa.europa.eu/publications/enisa-threat-landscape-2020>
- FBI INTERNET CRIME COMPLAINT CENTER (IC3). (2020). *2020 Internet Crime Report* [online]. [cited 2023-05-10]. Available at: <http://www.ic3.gov>
- FRANCO, A., MALHOTRA, N., & SIMONOVITS, G. (2014). Publication bias in the social sciences: Unlocking the file drawer. *Science*, 345(6203), pp. 1502–1505. <https://doi.org/10.1126/science.1255484>
- GOODE, E., & BEN-YEHUDA, N. (2009). *Moral panics: The social construction of deviance* (2nd ed.). Wiley-Blackwell.
- IBM SECURITY. (2021). *Cost of a Data Breach Report 2021* [online]. [cited 2023-05-10]. Available at: <https://www.ibm.com/security/data-breach>
- IOANNIDIS, J. P. A. (2005). Why most published research findings are false. *PLoS Medicine*, 2(8), e124. <https://doi.org/10.1371/journal.pmed.0020124>
- KASPERSKY LAB. (2019). *Global IT Security Risks Survey Report* [online]. [cited 2023-05-10]. Available at: <https://www.kaspersky.com>
- LEVI, M. (2017). Assessing the trends, scale and nature of cybercrime: overview and issues. *Crime, Law and Social Change*, 67(3), pp. 209–224.
- PONEMON INSTITUTE. (2023). *Cost of a Data Breach Report 2023* [online]. Sponsored by IBM Security. [cited 2023-05-10]. Available at: <https://www.ponemon.org>

STERNE, J. A., GAVAGHAN, D., & EGGER, M. (2001). Publication and related bias in meta-analysis: power of statistical tests and prevalence in the literature. *Journal of Clinical Epidemiology*, 53(11), pp. 1119–1129.

VAN EETEN, M., BAUER, J. M., ASGHARI, H., TABATABAIE, S., & WILSON, D. (2010). *The role of internet service providers in botnet mitigation: An empirical analysis based on spam data* [online]. OECD Publishing. [cited 2023-05-10].

WICKRAMASEKERA, R., & BALI, R. K. (2015). Cybercrime: A global problem. *Journal of Information Security*, 6(3), pp. 171–180.

WORLD BANK. (2024). *World Development Indicators* [online]. [cited 2023-05-10]. Available at: <https://data.worldbank.org>